Purva Chiniya

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Education

University of Maryland, College Park — College Park, MD
Master of Science in Computer Science — *Aug 2023 – May 2025*
Courses: Distributed Systems for ML, Advanced Numerical Optimization, LLM Security, ML at Scale

Indian Institute of Technology Roorkee — Roorkee, India
Bachelor of Technology in Electrical Engineering — *Aug 2017 – June 2021*
Awards: Best Project Award, Top Innovation Award

Experience

Amazon.com — Bellevue, WA
Applied Scientist, Amazon Ads — *June 2025 – Present*

- Architected a multi-agent autonomous framework for Sponsored Products using LLM-based reasoning agents to automate campaign optimizations for 1M+ advertisers; engineered scalable data pipelines in PySpark and AWS Glue to synthesize multi-terabyte seller and customer telemetry from S3 data lakes.
- Deployed a runtime safety model for Alexa+ using a vLLM inference engine handling 80M+ daily requests at 35ms P90 latency; optimized throughput utilizing continuous batching and PagedAttention.
- Directed the alignment strategy using Adversarial Fine-Tuning (AFT) on 7B+ models, boosting precision 8x (4% → 34%) and cutting False Positive Rate by 97% to effectively mitigate reasoning over-refusals.

Sentient Foundation — Remote, US
AI Research Intern — *Dec 2024 – May 2025*

- Built Open Deep Search, a reasoning agent achieving SOTA (75.3%) on FRAMES and outperforming GPT-4o Search Preview via adaptive planning and dynamic query rewriting for complex multi-hop retrieval.
- Built the CodeAct (Chain-of-Code Action) orchestration layer using SmolAgents and LiteLLM to execute multi-step reasoning; implemented a reproducible evaluation harness comparing ReAct vs. CodeAct.
- Developed an HPO hyperparameter optimisation fine-tuning pipeline for data-mixture and hyperparameter search, cutting tuning iterations by 30% and reducing fine-tuning cycle time by 40% across 3B–70B parameter models.

Amazon.com — Bellevue, WA
Applied Scientist Intern — *May 2024 – Aug 2024*

- Engineered a novel inference-time safety steering mechanism utilizing classifier gradients to guide decoding, achieving a 20% reduction in Attack Success Rate (ASR) on AdvBench while cutting false refusal rates by 50%.
- Benchmarked performance against SOTA defenses (SafeDecoding, DPO) on Llama2-7B, validating the method's effectiveness on jailbreak robustness; research accepted for publication at AMLC 2024 and LREC 2026.

Cisco Systems — Bangalore, India
Software Engineer II — *Aug 2021 – Aug 2023*

- Optimized C++ BGP routing modules for Google peering traffic, resolving critical concurrency bottlenecks to drive a 16% revenue increase for the XR Business Unit via enhanced service availability.

Research Publications

[1] **Gradient-Controlled Decoding: A Safety Guardrail for LLMs with Dual-Anchor Steering.** Accepted at LREC 2026.

[2] **SecRepoBench: Benchmarking Secure Code Generation in Real-World Repositories.** Repo-level vuln benchmark spanning 27 C/C++ repositories (318 tasks); revealed a 31% security drop vs single-file baselines. Accepted at LLM4Code 2026. arXiv:2504.21205.

[3] **Open Deep Search: SOTA Agentic Search via Planning.** Achieved 75.3% on FRAMES and outperformed GPT-4o Search Preview by +9.7% via adaptive planning and tool use. Preprint (2025). arXiv:2503.20201.

[4] **AV-RIR: Audio-Visual Room Impulse Response Estimation.** Multi-task learning fusing audio-visual cues to reconstruct acoustic environments. Accepted at CVPR 2024. arXiv:2312.00834.

[5] **CoSyn: Implicit Hate Speech Detection via Hyperbolic Networks.** Hyperbolic GNN (GCN) + TreeLSTM to model latent social dynamics in conversational threads. Accepted at EMNLP 2023. arXiv:2303.03387.

[6] **LipGER: Visually-Conditioned ASR Error Correction.** Visual modality pipeline to correct ASR errors in noise-heavy environments and improve robustness. Accepted at INTERSPEECH 2024. arXiv:2406.04432.

Technical Skills

Languages: Python, Java, C++, SQL, Bash
ML/Systems & Tools: PyTorch, Triton, vLLM, DeepSpeed, FlashAttention, Codex, Git, Docker, Cursor, Claude Code